

X-RAY FLUORESCENCE DATA ANALYSIS USING A BLANK CORRECTION APPROACH

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ABSTRACT

Due to overlap of x-ray emission lines in energy dispersive x-ray detectors, x-ray fluorescence (XRF) data from x-ray microprobe and nanoprobes measurements must be fitted to obtain reliable elemental maps. To do this, many programs initially remove a per-pixel baseline prior to fitting to account for a background XRF continuum. Previously, we demonstrated an alternative background correction where in the average blank spectrum is calculated from non-sample pixels and subtracted from the raw XRF data prior to fitting. By comparison to blank correction, we demonstrated that per-pixel baseline correction can introduce artifacts in the apparent concentration of dilute elements, particularly when there are strong fluorescence peaks nearby, showed that this is important under biologically relevant conditions, and that such artifacts incurred significant, systematic errors in quantitation.

M-BLANK, a program written in MATLAB and the foundational aspect of that work, implements this blank correction approach. In all comparisons performed between M-BLANK with programs that utilize per-pixel baseline correction, very similar resulting artifacts were observed. The author specifically notes that these findings result from the use of per-pixel baseline correction and not any fault with the associated architecture of the software.

This paper will discuss the M-BLANK program, features, functionality and development challenges which enable the use of a blank subtraction approach for spectrum fitting. M-BLANK is a MATLAB program, capable of using many different types of background functions, but is most powerful at employing the blank correction approach.

INTRODUCTION

Elemental mapping in cells, tissues, and other relevant biological samples using XRF at third-generation (and newer) synchrotron sources has become a very common and powerful tool for characterizing the biological metallome with resolution limits of tens of nanometers (Carmona *et al.*, 2019, Jin *et al.*, 2017, Ortega *et al.*, 2009, Paunesku *et al.*, 2009, Penner-Hahn, 2014, Punshon *et al.*, 2013, Pushie *et al.*, 2014, Vogt and Ralle, 2013, West *et al.*, 2014, Zhao *et al.*, 2014). XRF detected flow cytometry has also been developed (Crawford *et al.*, 2016, Crawford and Penner-Hahn, 2018). To a first approximation, XRF is the only approach capable of determining total elemental content independent of the chemical form and environment. As such, XRF is an ideal

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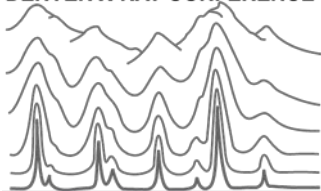
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tool for interrogating the cellular metallome. However, due to overlap of emissions caused by the less than perfect detector energy resolution of energy dispersive x-ray detectors, XRF data must be fitted to obtain quantitative results.

Many programs exist for the fitting of XRF data to obtain quantitative elemental compositions, such as AXIL (Janssens *et al.*, 1996), GeoPIXE/Dynamic Analysis (Ryan & Jamieson, 1993, Ryan *et al.*, 1995), MAPS (Vogt, 2003), PyMCA (Solé *et al.*, 2007), PyXRF (Li *et al.*, 2017), M-BLANK (Crawford *et al.*, 2019), and SMAK (Webb, 2011) which utilizes an embedded version of PyMCA. Each program uses some formulation of a background function to correct data prior to fitting; and for some the use of per-pixel baselines is obligatory. Recently however, it was shown that not all background corrections are equivalent and that, under certain conditions, background correction, using per-pixel baselines, can lead to significant errors in quantitation (Crawford *et al.*, 2019).

Ideally, calculated baselines used for correction in quantitative XRF using monochromatic sources should be mathematically orthogonal to signal details; unfortunately, they are not, and even small changes in elemental composition of the signal can lead to significantly different baselines, both in morphology and amplitude. This variability in calculated baselines is obvious as baselines for blank pixels are systematically lower than those for cell pixels; so much so, that baseline amplitudes alone can give a very good reproduction of where the cell is. As such, the utilization of per-pixel baselines risks the incorporation of systematic error which have been shown to result in sample specific apparent changes in elemental quantitation ranging from +10% to -200%.

Baseline variation is sample-specific and has been found to be most severe in spectral regions of high counting statistics when there are two adjacent peaks with significantly different amplitudes; especially when a very weak emission rests on the low energy shoulder of a very high intensity fluctuating emission. Most importantly, because this artifact is sample-specific, it is incredibly difficult to predict. For example, Crawford *et al.*, (2019) demonstrated erroneous Cl quantitation caused by intense K emission in yeast cells, while in glial cells intense exogenous Cl emissions led to erroneous S concentrations. Although the specific error was different and elementally unique for both cases, the root cause was the same; sharp change in the intensity of a neighboring emission resulted in the calculated baseline overcorrecting the area underneath the curve for the weaker emission. These examples both resulted from the distortions that occur when two adjacent elements have very different signal intensities.

Although it was obvious for the Cl and K, or Cl and S in the referenced example using yeast and glial cells, problems arising from baseline distortion were (and remain) far more general and reaching. This is because per-pixel baseline morphology is inherently context dependent in a way that gives rise to baselines that vary both in amplitude and shape. As such, this type of artifact decreases quantitative precision and makes it difficult to compare quantitation within a single image, across multiple images within an experimental group, and across multiple images across multiple experimental groups.

Ultimately, these errors were noted by comparing the blank-corrected answers from M-BLANK to the per-pixel corrected answers from the same data using baselines calculated from MAPS,

PyMCA, and PyXRF (but fitted in M-BLANK to avoid potential differences in results that could arise due to differences in software architecture). In all three comparisons, correction by the per-pixel baselines introduced very similar non-negligible differences in the apparent concentrations of all the elements; and, most notably when a trace element emission rested on the low energy shoulder of a fluctuating high intensity fluorescence peak, the noted difference was catastrophic. Although these errors in quantitation from per-pixel baseline correction were shown to be consequential under biologically relevant conditions, they are to be expected in any sample with significantly changing concentrations. As such, caution should be exercised for any samples in which XRF data is corrected using per-pixel baselines. This is especially true when the data is of utmost importance and refitting of data is not readily feasible, e.g., preliminary fitting to be done onboard the Mars 2020 Rover's Planetary Instrument for X-ray Lithochemistry (PIXL, <https://mars.nasa.gov/mars2020/mission/instruments/pixl/>) due to limited data bandwidth prior to return of the fitted values to Earth.

The objective of this manuscript is to discuss the M-BLANK software package (Crawford, 2019), which exists as the foundation of the above referenced work, to provide an overview of its capabilities and functionality; specifically the details of how blanks are calculated. M-BLANK is a MATLAB program for the fitting of XRF data capable of using many different types of background functions, most notably the blank correction approach. For further detail on fitting and handling of backgrounds and calculating blanks, and the observations summarized herein, please see Crawford *et al.*, 2019.

EXPERIMENTAL

Source Code

M-BLANK is compatible with Mac, UNIX, and PC, and at time of writing is available for download as both compiled binaries and source code (Crawford, 2019).

It was most recently compiled using MATLAB 2017a. The compiled code can be downloaded and installed without the need of a MATLAB License. To view and run the source code, end users will need MATLAB as well as the following toolboxes installed:

1. Communications Toolbox, v7.2
2. Control System Toolbox, v10.2
3. Curve Fitting Toolbox, v3.5.2
4. DSP System Toolbox, v9.4
5. Image Processing Toolbox, v10.0
6. MATLAB Compiler, v6.4
7. Optimization Toolbox, v7.6
8. Parallel Computing Toolbox, v6.10
9. Robust Control Toolbox, v6.3
10. Signal Processing Toolbox, v7.4
11. Statistics and Machine Learning Toolbox, v11.1

12. Symbolic Math Toolbox, v7.2

Compatibility and Functionality

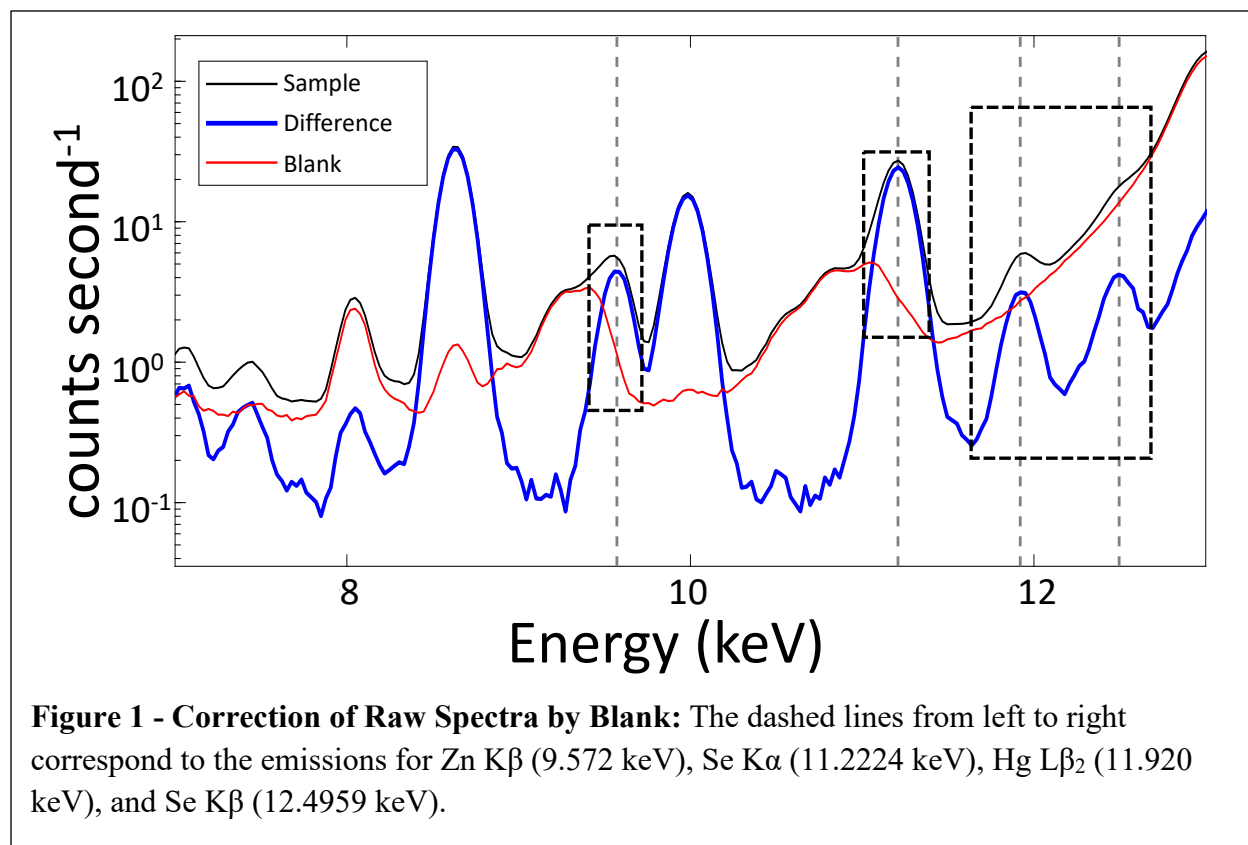
M-BLANK is compatible with various data formats from many synchrotrons including: **Advanced Photon Source** (Sector 2: *.mda, *.mca, *.h5; Bionanoprobe: *.nc, *.mda; Sector 20: *.hdf5, *.mda); **Stanford Synchrotron Radiation Lightsource** (all imaging beamlines: *.tar.gz, *.mca [legacy, format prior to Xpress3]); **Canadian Light Source** (VESPERS; SXRMB; IDEAS; SGM: *.hdf5, *.nc [legacy]); **Australian Synchrotron** (XFM BioMAIA blog format); **Diamond** (B16, I18; both are currently in beta testing); and the **European Synchrotron Radiation Facility** (ID13; currently in beta testing).

M-BLANK is capable of exporting fitted data to both Excel and SMAK (Webb, 2011) compatible ASCII and hdf5 formats. Users are also able to export designated map regions of interest as well as associated statistics.

The earliest versions of M-BLANK relied solely on the X-Ray Data Booklet (Thompson *et al.*, 2009) for absorption and emission energies. Unfortunately, that reference is not exhaustive, e.g., it does not contain references for L1 absorption edges and only contains references for M5 edges of the M series. Current versions of M-BLANK have incorporated the xraylib library for X-ray–matter interactions from ESRF (Schoonjans *et al.*, 2011). Future versions will be switched over entirely.

Parameterization

Fitting parameters (e.g., detector bin-to-energy calibration, detector response function, emission peak shapes, Compton (inelastic) scatter profile, Raleigh (elastic) scatter profile, etc.) can be derived from both samples and thin foil standards used for mass calibration. Because sample scans can be partitioned into ‘sample’ and ‘blank’ pixels, M-BLANK can use the difference spectrum from the mean of both regions for parameterization. As can be seen in **Figure 1**, blank correction of the sample spectra, removes more than just a baseline XRF continuum; the blank contains structure and non-sample peak information. One of the advantages of correcting by a blank (rather than a baseline) is that the removal of this structure yields better precision and accuracy for fitting parameters as it changes, subtly, the apparent centroids of some of the fluorescence emissions. This can be seen by looking at the dashed black boxes which highlight, from left to right, emissions corresponding to Zn K β (9.572 keV), Se K α (11.208.4 keV), Hg L β_2 (11.920 keV), and Se K β (12.4959 keV). Upon inspection it can be noted that the apparent centroids for Zn K β , Se K α , and Se K β shift slightly to higher energies, whereas the apparent centroid for Hg L β_2 is slightly shifted to a lower energy.



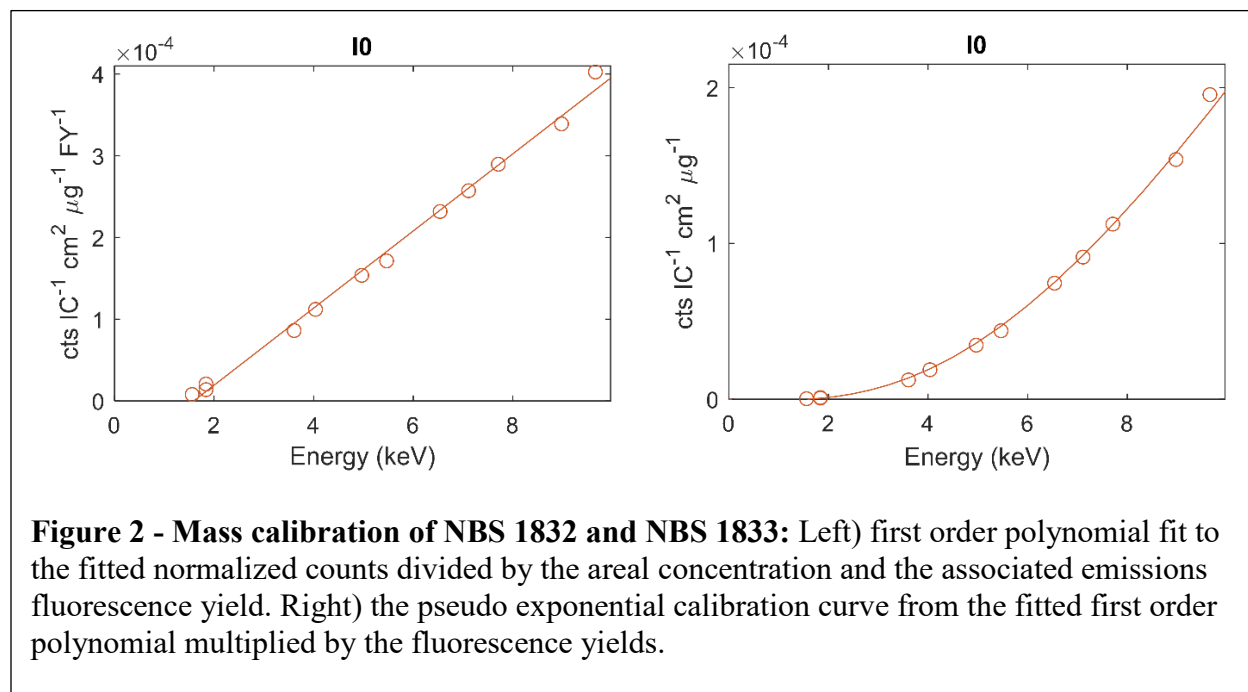
Mass Calibration

M-BLANK comes equipped with the standard information to readily apply mass calibrations from APS, Sector 2 (NBS1832 and NBS1833), as well as most of the thin foil MICROMATTER™ standards used for mass calibration at SSRL. M-BLANK also accommodates end-user specific standards by allowing the user to create their own standard reference files. The current version of M-BLANK fits, as a function of emission energy, a first-order polynomial to the ratio of fitted normalized counts (counts divided by the upstream ion chamber) divided by the areal concentration ($\mu\text{g cm}^{-2}$ reference standard listed value) and theoretical fluorescence yield using xraylib (Schoonjans *et al.*, 2011). This fluorescence-normalized calibration curve from the first-order polynomial is then multiplied by the associated element's fluorescence yield to create a pseudo exponential calibration curve. For an example of mass calibration using NBS1832 and NBS1833, see **Figure 2**.

DISCUSSION

Background correction in M-BLANK is versatile and robust as M-BLANK is capable of calculating both baselines and blanks. If a blank is not possible and a baseline is to be used, it is highly recommended that a constant (rather than per-pixel) baseline be calculated across the entire

dataset, as the use of a constant baseline avoids the introduction of systematic errors arising from the context-dependent nature of per-pixel baselines.



Proper blank calculation is an iterative process. *A priori* knowledge of each pixel's identity is necessary. Crude binning and intensity filtering aids in the initial assignment of a blank; however, issues such as background contamination (e.g., crystallized chloride salts) or heterogeneity in the elemental signature of the template material (e.g., Si fluorescence from a Si_3N_4 wafer) can pose real challenges to proper background correction. Once blanks are calculated for individual images, the low count rate nature of synchrotron x-ray fluorescence imaging can result in very noisy blanks; however, increased precision is possible. The blanks can be compared across multiple images to identify samples with identical backgrounds and the spectra from all the background samples can be combined to calculate a global blank.

An additional caveat involves the time sensitive nature of synchrotron x-ray fluorescence imaging data collection. Because of time constraints during data collection, choice cells or tissue regions are typically selected from sample slides in advance and then scanned individually. This results in multiple individual images that are part of a larger overall image. Analysis can be simplified if these images are stitched (or pieced) back together. Doing so allows better estimation of a blank and can facilitate faster data analysis and processing. Additionally, this also makes possible enhanced accuracy and precision when registering XRF elemental maps with classical histology using stains or fluorescent probes (e.g., DAPI labeled nuclei or GFP labeled vacuoles). Unfortunately, piecing together separate parts of an overall larger map is not always straight forward since the grid coordinates of the image don't always align perfectly.

Stitching Images

As mentioned above, multiple scanned regions of an overall larger image can be stitched together – a “patchwork quilt” of scans so to speak – quite easily in M-BLANK. Using the reported x- and y-coordinates of the stepper motors used during data collection, M-BLANK maps the spectra and all associated scalars into a common coordinate grid. Unfortunately, precision microstages do not deal in integer coordinates and the associated grid coordinates from multiple scans can be offset by up to 50% of the associated pixel step size. For example, if image 1 is collected at x coordinates ranging from 0 to 20 with a step size of 1, and image 2 is collected at x coordinates ranging from 0.1 to 20.1 with a step size of 1, then the offset between the two images will be 10%. To account for such an offset, data can either be interpolated or shifted (without changing any values) into a common coordinate system. Although it might seem best to interpolate using a cubic spline interpolant to move from one coordinate system into another, this is not the case with x-ray fluorescence imaging data since the spectra are typical sparsely populated; this is especially true for most biological samples, where signal intensity is dominated by low count statistics regions across the spectra. Sparse data is characterized by being dominated by zeros, i.e., the absence of signal, and when such data is interpolated, the interpolated values are almost always shifted closer to zero. Though not shown herein, initial studies in M-BLANK using interpolative stitching resulted in datasets with apparent signals that had been reduced (relative to the signal prior to interpolation) by as much as 40%. As such, it is best to shift the spectra (rather than interpolate), unadulterated, using the minimal offset that best aligns all coordinates from all images into a common grid.

Identification and Refinement of ‘blank’ pixels

For images containing both ‘sample’ and ‘blank’ pixels (e.g., any type of cell sparsely plated on a Si_3N_4 wafer), ‘blank’ pixels can be automatically partitioned in an iterative process using user-designated emission lines (or ROIs centered thereon) (Crawford, 2015, Crawford *et al.*, 2016, Crawford & Penner-Hahn, 2018, Crawford *et al.*, 2018). Initially, all pixels with intensities greater than a user-set threshold (typically, 0.1-0.2 the maximum emission or ROI signal) become defined as ‘non-blank’. Each iteration then calculates the mean, $\bar{x}$, and standard deviation, σ , of the remaining ‘blank’ pixels. All pixels with intensities greater than $\bar{x} + m\sigma$, where m is a user-defined multiple of σ (typically, 2), are redefined as ‘sample’. This process repeats until the populations become self-consistent.

The population of ‘blank’ pixels can be further refined for calculating the blank spectrum by identifying and rejecting ‘abnormal’ pixels, e.g., pixels that contain contaminants such as dust. Similar to identifying ‘blank’ pixels above, the histograms of all the fitted elements from all ‘blank’ pixels are fitted to a normal distribution yielding a fitted standard deviation, σ , and population mean, $\bar{x}$. All pixels with any elemental intensity I such that $|I - \bar{x}| > m\sigma$, where m is a user defined multiple of σ (typically, 4), is rejected from the population and the process repeated until the population of identified pixels is self-consistent.

Calculating, Recalculating, and Combining Blanks

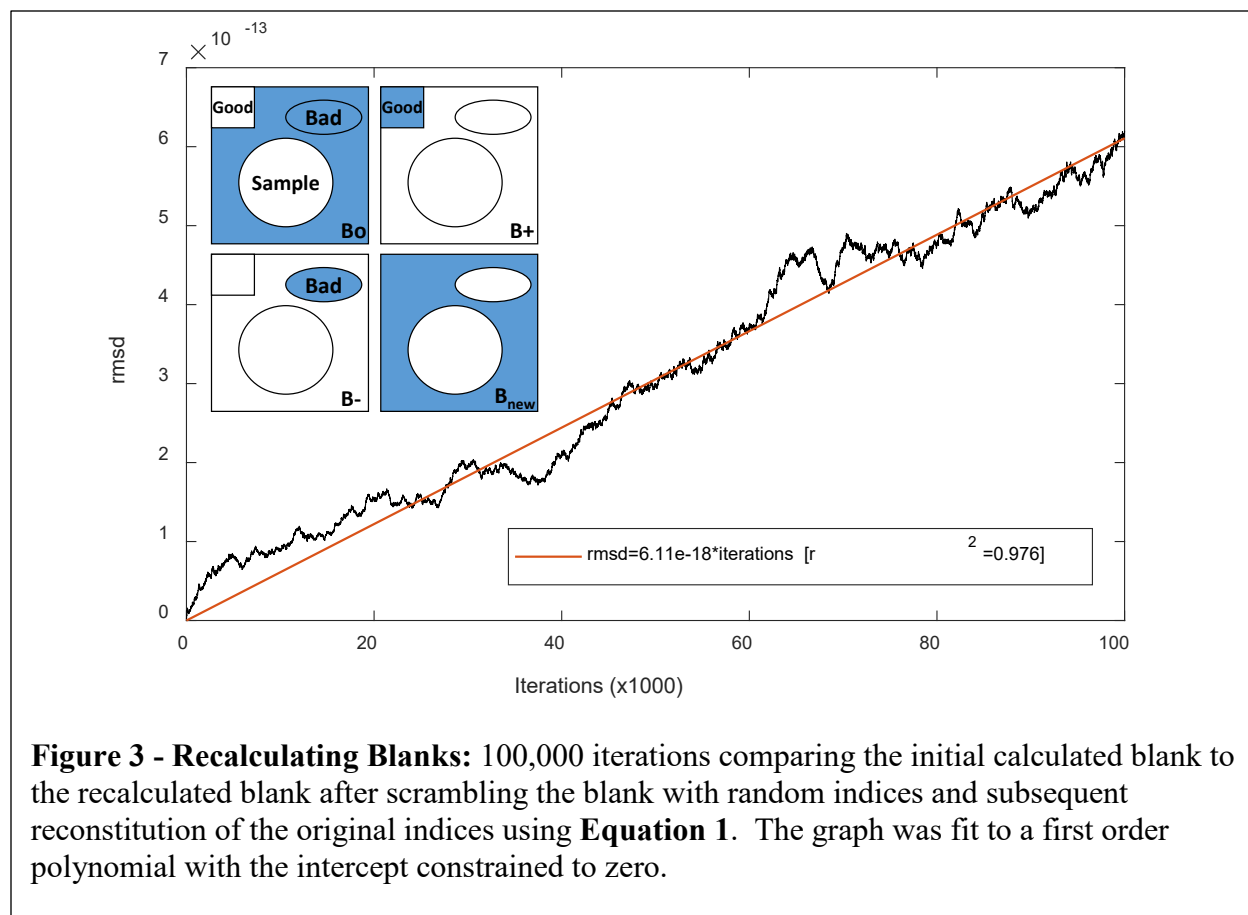
A blank can be calculated from the mean of all blank pixels (see: *Identification and Refinement of 'blank' pixels*, above) in the associated scan or from another scan collected under identical conditions (e.g., sample template, sample prep, collection geometry, detector, detector settings, etc.). Once calculated, subsequent calculations use a matrix algebra approach to minimize the amount of time necessary for recalculation (see below). Users are able to compare across multiple blanks from multiple samples allowing for fine-tuning of the blanks. Blanks can then be combined into a single “master blank” to yield superior signal-to-noise of the blank and this “master blank” can then be applied across multiple scans for improved precision.

All initial blanks are calculated by reading in the spectra from all background pixels and calculating the mean. After the initial blank has been calculated, there are two ways that future blanks can be calculated (e.g., if the blank is optimized by removing contamination from the background, see: *Identification and Refinement of 'blank' pixels*). One way is to read in the spectra from all the background pixels and recalculate the blank by taking the mean (this is identical to the way the initial blank is calculated). Alternatively, all subsequent blanks can be calculated as a linear combination of the current calculated blank, the spectral mean of the pixels that need to be removed (included in the previous blank calculation but now need to be removed), and the spectral mean of the pixels that were not included and now need to be included. This second approach is much faster as it avoids the need to read in the spectra from all the background pixels. As such, for any particular dataset, M-BLANK calculates a weighted mean using linear algebra for all subsequent calculations, such that

$$B_{new} = \frac{\begin{bmatrix} B_0 & B_+ & B_- \\ \vdots & \vdots & \vdots \\ n & n & n \end{bmatrix} \times \begin{bmatrix} +N_0 \\ +N_+ \\ -N_- \end{bmatrix}}{\sum N}$$

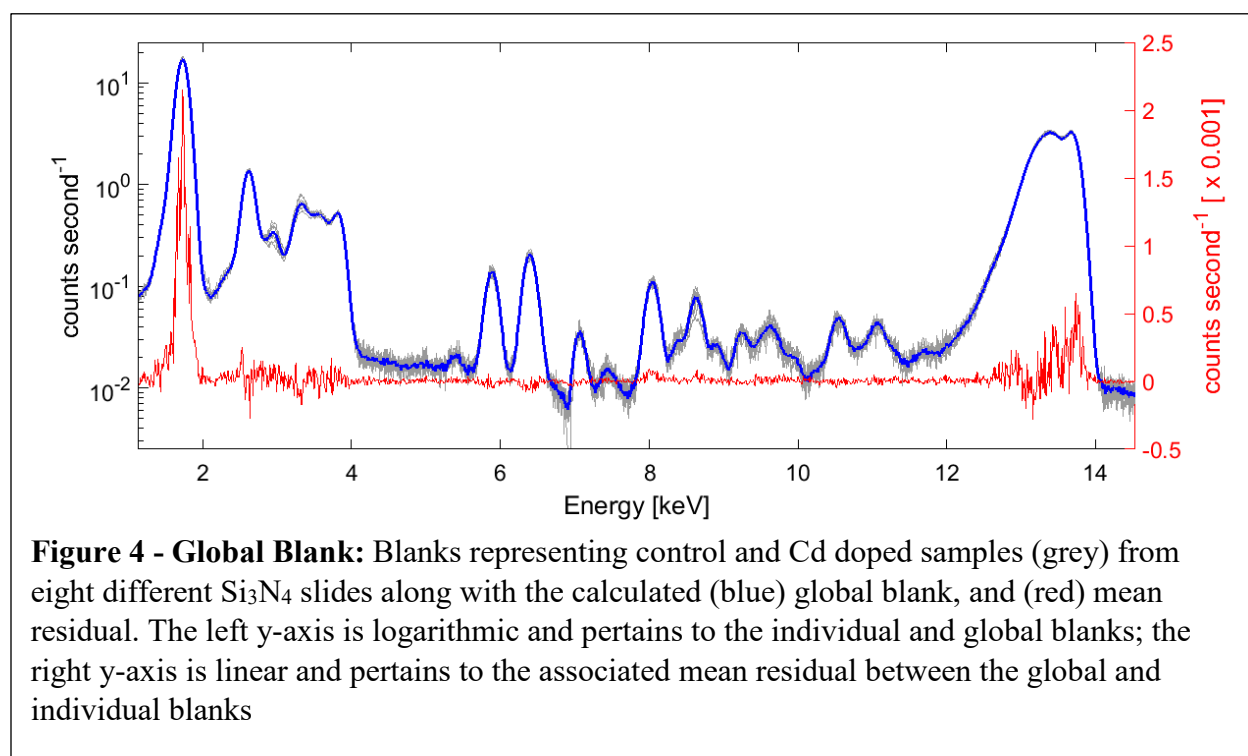
Equation 1

where, B_{new} is the blank to be calculated; B_0 , B_+ , and B_- are column vectors corresponding the current blank, the mean spectra (or blank) from the new pixels to be added, and the mean spectra (or blank) from the pixels in the current blank that are being removed, respectively; matrix N is a column vector of coefficients $+N_0$, $+N_+$, and $-N_-$, corresponding to the number of pixels used for calculating B_0 , B_+ , respectively. An example of B_0 , B_+ , B_- , and B_{new} is schematically illustrated in the inset of **Figure 3**, where N_0 , N_+ , N_- , and $\sum N$ would be the total number of blue pixels in B_0 , B_+ , B_- , and B_{new} , respectively.



For small datasets, the difference in time is negligible, however for very large datasets with very many pixels the difference can be significant. The viability of this approach was tested iteratively. To do this, an initial blank was calculated, B_{initial} . Then, each iteration, n , involved two steps. Step one calculated a new (nonsensical) blank by pseudo-randomly reassigning the population of identified blank pixels. This new population was then used to generate B_+ and B_- which were combined with B_{n-1} using **Equation 1**. Step two then used **Equation 1** with the original indices to recalculate the blank, B_n . For each iteration the root-mean-squared-deviation (RMSD) between B_{initial} and B_n was calculated. After 100,000 iterations the initial and final blank were identical within computational accuracy, having an associated RMSD of less than $1\text{E-}12$. **Figure 3** plots the total incorporated error (or RMSD) as a function of the total number of times the blank had been recalculated using linear algebra rather than explicitly loading the spectra from every pixel to calculate the mean. Based on a first order fit to that plot, $8\text{E}13$ calculations are required before a typical digitized spectrum (2^{11} bins) will contain a total root-summed-squared-deviation of one count across the spectra. As such, performing a blank calculation each second, this would take 2.5 million years. Therefore, the matrix algebra approach to blank calculation provides a very fast way to recalculate blanks with no practical introduction of error.

The individual blanks calculated from each image can be combined with images collected under identical conditions. This can provide for enhanced signal-to-noise, and therefore precision, when correcting by a blank. Additionally, the applicability of blanks to other scans can be very advantageous as it can allow for blank correction of an image that contains no non-sample pixels. Different from previous versions of M-BLANK, the most recent versions of M-BLANK store the associated fitted scalars (elements and scatter), pixel indices, and ratios for the various normalization terms, e.g., the upstream and downstream ion chambers. The associated pixels from the global set of ‘blank’ pixels can be filtered (see: *Identification and Refinement of ‘blank’ pixels*) and combined using a weighted mean approach similar to **Equation 1**. The applicability of this approach is demonstrated in **Figure 4** which shows the overlay of blanks from eight different Si₃N₄ slides, the combined mean blank, and the mean residual between the eight individual blanks and the combined blank.



Data Processing and Fitting

M-BLANK makes data processing and fitting quick and simple. Batch fitting is possible, and for most samples, M-BLANK is capable of batch processing including automatic pixel partitioning into ‘sample’ and ‘blank’ pixels, ‘blank’ pixel refinement to automatically identify and reject abnormal/anomalous pixels, calculating the blank spectrum, fitting, thickness normalization (if blank pixels are present), and subsequent export of fitted data formats. Furthermore, reprocessing is simple. For example, if a mass calibration curve is modified or updated, or a global blank is updated, or ion chambers are reassigned, or fitting parameters are updated or a multitude of any other things, reprocessing data is simple and can be done in batch mode.

CONCLUSIONS

M-BLANK is a program for the fitting of x-ray fluorescence datasets that can take advantage of the blank approach. The most compelling reason to use the blank approach is that it avoids the introduction of systematic errors which are known to occur when data is corrected by per-pixel baselines due to their context-dependent nature.

Calculating an appropriate blank for x-ray fluorescence data correction is typically an iterative process. M-BLANK, which is engineered specifically to address this need, takes advantage of using a blank both for generating more accurate fitting parameters and for providing more accurate and precise answers.

The ability to combine blanks from multiple samples and the ability to borrow a blank from a different sample to correct an image in which there are no background pixels makes the use of the blank correction approach highly desirable for most biological applications using x-ray fluorescence imaging to investigate, among other things, the biological metallome.

M-BLANK is licensed under a Creative Commons Attribution-Non-Commercial-ShareAlike 4.0 International License, <https://creativecommons.org/licenses/by-nc-sa/4.0/>, and is available online as compiled binaries and source code (Crawford, 2019).

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